Soluzione Esercizi 1.5.2 algebra di Boole

1 Funzione a 2 variabili

A	B	F	G	H	I
0	0	1	1	1	1
0	1	0	1	0	1
1	0	1	1	1	0
1	1	1	1	0	0

Svolgimento delle Espressioni Logiche

$$F(A,B) = A\cdot\overline{\bar{A}B}+\bar{B} \iff A(A+\bar{B})+\bar{B} \iff AA+A\bar{B}+\bar{B} \iff A+A\bar{B}+\bar{B} \iff A(1+\bar{B})+\bar{B} \iff \boxed{A+\bar{B}}$$

$$G(A,B) = \bar{A}+B+A\bar{B}+\overline{A+\bar{B}} \iff \bar{A}+B+A\bar{B}+\bar{A}B$$

$$\iff \bar{A}(1+B)+B+A\bar{B} \iff \bar{A}+\underbrace{B+A\bar{B}}_{Distributiva} \iff \bar{A}+(B+A)\cdot(B+\bar{B}) \iff \bar{A}+B+A \iff 1+B \iff \boxed{1}$$

$$H(A,B) = \overline{\overline{\bar{A}\bar{\bar{B}}}B} \iff \overline{(\bar{A}+B)B} \iff \overline{\bar{A}B+B} \iff \overline{B(1+\bar{A})} \iff \boxed{\bar{B}}$$

$$I(A,B) = \bar{A}+\overline{A+\overline{AB}} \iff \bar{A}+\overline{A+\bar{A}+\bar{B}} \iff \bar{A}+\overline{(1+\bar{B})} \iff \bar{A}+\bar{1} \iff \bar{A}+0 \iff \boxed{\bar{A}}$$

2 Funzioni a 3 variabili

A	**B**	**C**	**F**	**G**	**H**	**I**	**L**	**M**
0	0	0	0	0	0	1	0	0
0	0	1	0	0	0	1	0	1
0	1	0	1	0	0	1	0	1
0	1	1	1	0	0	0	0	1
1	0	0	1	1	1	1	0	1
1	0	1	1	0	1	1	1	1
1	1	0	1	0	1	1	1	1
1	1	1	1	0	1	1	1	1

Svolgimento delle Espressioni Logiche

$$F(A,B,C) = CB+A+B \iff B(C+1)+A \iff B+A \iff \boxed{A+B}$$

$$G(A,B,C) = A\cdot A\bar{B}\cdot\overline{A\bar{B}C} \iff A\bar{B}\cdot(\bar{A}+B+\bar{C}) \iff \underbrace{A\bar{B}\bar{A}}_{0}+\underbrace{A\bar{B}B}_{0}+A\bar{B}\bar{C} \iff \boxed{A\bar{B}\bar{C}}$$

$$H(A,B,C) = \overline{\bar{A}+\overline{A+CB}A} \iff \overline{\bar{A}+\bar{A}\cdot\overline{CB}A} \iff \overline{\bar{A}+\underbrace{\bar{A}\cdot\bar{C}\bar{B}\cdot A}_{0}} \iff \bar{\bar{A}} \iff \boxed{A}$$

$$I(A,B,C) = AC+A\bar{B}+\overline{C(A+B)} \iff AC+A\bar{B}+\overline{CA+CB}$$

$$\iff AC+A\bar{B}+(\bar{C}+\bar{A})(\bar{C}+\bar{B}) \iff AC+A\bar{B}+\underbrace{\bar{C}\bar{C}}_{\bar{C}}+\bar{C}\bar{B}+\bar{A}\bar{C}+\bar{A}\bar{B}$$

$$\iff AC+\underline{A\bar{B}}+\underline{\bar{C}}+\underline{\bar{C}\bar{B}}+\bar{A}\bar{C}+\underline{\bar{A}\bar{B}} \iff \bar{B}\underbrace{(A+\bar{A})}_{1}+\bar{C}\underbrace{(1+\bar{B})}_{1}+AC+\bar{A}\bar{C}$$

$$\iff \bar{B}+\underline{\bar{C}}+AC+\underline{\bar{A}\bar{C}} \iff \bar{C}\underbrace{(1+\bar{A})}_{1}+\bar{B}+AC \iff \bar{C}+\bar{B}+AC$$

$$\iff \bar{B}+(\bar{C}+A)\underbrace{(\bar{C}+C)}_{1} \iff \boxed{\bar{B}+\bar{C}+A}$$

$$L(A,B,C) = (A+\bar{B})(A+B)(\bar{A}+B+C) \iff (A+\underbrace{B\cdot\bar{B}}_{0})(\bar{A}+B+C)$$

$$\iff A\bar{A}+AB+AC \iff BA+AC \iff \boxed{A(B+C)}$$

$$
\begin{aligned}
&M(A,B,C) = (A+B+C)(\bar{A}+\bar{B}+\bar{C})+B \\
&\iff \underbrace{A\bar{A}}_{0} + A\bar{B} + A\bar{C} + B\bar{A} + \underbrace{B\bar{B}}_{0} + B\bar{C} + C\bar{A} + C\bar{B} + \underbrace{C\bar{C}}_{0} + B \\
&\iff A\bar{B} + A\bar{C} + B\bar{A} + \underline{B\bar{C}} + C\bar{A} + C\bar{B} + \underline{B} \\
&\iff B(1+\bar{C}) + A\bar{B} + A\bar{C} + B\bar{A} + C\bar{A} + C\bar{B} \iff \underline{B} + A\bar{B} + A\bar{C} + \underline{B\bar{A}} + C\bar{A} + C\bar{B} \\
&\iff B + A\bar{B} + A\bar{C} + C\bar{A} + C\bar{B} \iff \underbrace{(B+\bar{B})}_{1}(B+A) + A\bar{C} + C\bar{A} + C\bar{B} \\
&\iff A + B + A\bar{C} + C\bar{A} + C\bar{B} \iff A + B + A\bar{C} + C\bar{A} \\
&\iff \underbrace{(A+\bar{A})}_{1}(A+C) + \underbrace{(B+C)}_{1}(B+\bar{C}) \iff A + C + B + \bar{C} \iff \boxed{A+B+C}
\end{aligned}
$$

3 Funzioni a 4 variabili

A	**B**	**C**	**D**	**F**	**G**	**I**	**AB**	**BC**	**AC**	**(AB+BC+AC)**	**H**
0	0	0	0	0	1	0	0	0	0	0	**0**
0	0	0	1	0	1	0	0	0	0	0	**0**
0	0	1	0	1	1	1	0	0	0	0	**0**
0	0	1	1	1	1	1	0	0	0	0	**0**
0	1	0	0	0	1	0	0	0	0	0	**0**
0	1	0	1	0	1	0	0	0	0	0	**0**
0	1	1	0	1	1	1	0	1	0	1	**0**
0	1	1	1	1	0	1	0	1	0	1	**1**
1	0	0	0	1	1	1	0	0	0	0	**0**
1	0	0	1	1	1	1	0	0	0	0	**0**
1	0	1	0	1	1	1	0	0	1	1	**0**
1	0	1	1	1	1	1	0	0	1	1	**1**
1	1	0	0	0	1	1	1	0	0	1	**0**
1	1	0	1	0	1	1	1	0	0	1	**1**
1	1	1	0	1	1	1	1	1	1	1	**0**
1	1	1	1	1	1	1	1	1	1	1	**1**

Svolgimento delle Espressioni Logiche

$$
\begin{aligned}
&F(A,B,C,D) = CD + A\bar{B} + \overline{A\bar{B} + \bar{A}} + C\bar{D} \iff \\
&\iff CD + A\bar{B} + A(\overline{\bar{B}A}) + C\bar{D} \iff \boxed{CD + A\bar{B} + A\bar{B} + C\bar{D}} \\
&\iff C(D+\bar{D}) + A\bar{B} + A\bar{B} \iff C + A\bar{B} + A\bar{B} \iff \boxed{C + A\bar{B}}
\end{aligned}
$$

$$
\begin{aligned}
&G(A,B,C,D) = \bar{B} + AB + \overline{CD} + \bar{C} \iff (\bar{B}+A)\underbrace{(\bar{B}+B)}_{1} + (\bar{C}+C)\underbrace{(\bar{D}+\bar{C})}_{1} \iff \\
&\iff \boxed{\bar{B} + A + \bar{D} + \bar{C}}
\end{aligned}
$$

$$
\begin{aligned}
&H(A,B,C,D) = \bar{A}BCD + AB\bar{C}D + A\bar{B}CD + ABCD \iff \\
&\iff BCD\underbrace{(A+\bar{A})}_{1} + A\bar{B}CD + AB\bar{C}D \iff BCD + A\bar{B}CD + AB\bar{C}D \iff \\
&\iff BD\ \underbrace{(C + A\bar{C})}_{\substack{(C+\bar{C})(A+C) \\ 1}} + A\bar{B}CD \iff BD(A+C) + A\bar{B}CD \iff BDA + BDC + A\bar{B}CD \iff \\
&\iff D(BA + \underbrace{BC + A\bar{B}C}_{C(B+A\bar{B})}) \iff D(BA + C\ \underbrace{(B + A\bar{B})}_{\substack{(B+A)(\bar{B}+B) \\ 1}}\ ) \iff \\
&\iff D(BA + C(B+A)) \iff D(BA + CB + CA) \iff \\
&\iff \boxed{D(B(A+C)+AC)} \vee \boxed{D(BA+BC+AC)} \qquad\qquad I(A,B,C,D) = \boxed{A+C}
\end{aligned}
$$